

## Geometric patterns



1

- a 3
- b The number of matchsticks is equal to three times the number of triangles.
- c  $M = 3T$

2

a

|                 |   |   |   |   |
|-----------------|---|---|---|---|
| Figure number   | 1 | 2 | 3 | 4 |
| No. matchsticks | 5 | 6 | 7 | 8 |

- b The number of matchsticks needed to construct a shape is equal to the entry number of that shape plus four.
- c  $M = F + 4$

3

- a 3
- b The number of matchsticks increases by three
- c The number of matchsticks is equal to three times the number of squares plus one.
- d  $M = 3S + 1$

4

a

|                           |   |   |   |    |
|---------------------------|---|---|---|----|
| Entry number ( $E$ )      | 1 | 2 | 3 | 4  |
| No. of tiles used ( $T$ ) | 1 | 4 | 9 | 16 |

- b The number of tiles is equal to the entry number multiplied by itself.
- c  $T = E^2$

## Equations and tables

5

- a  $y = x - 6$
- b  $y = 3x$
- c  $y = 2x + 5$

6

a

- i The value for  $y$  is equal to the value for  $x$  plus eight.

ii  $y = x + 8$



**b**

i The value for  $y$  is equal to the value for  $x$  divided by three.

ii  $y = x \div 3$

**c**

i The value for  $y$  is equal to the value for  $x$  minus five.

ii  $y = x - 5$

**d**

i The value for  $y$  is equal to the value for  $x$  times seven.

ii  $y = 7x$

**e**

i The value for  $y$  is equal to the value for  $x$  times five plus two.

ii  $y = 5x + 2$

**f**

i The value for  $y$  is equal to the value for  $x$  times minus three plus seventeen.

ii  $y = -3x + 17$

**7**

**a**  $y = x - 3$

**b**  $y = 5x$

**c**  $y = x + 7$

**d**  $y = 2x + 2$

**e**  $y = 2x - 8$

**f**  $y = \frac{x}{9}$

**8**

**a**

|     |   |    |    |    |
|-----|---|----|----|----|
| $x$ | 1 | 2  | 3  | 4  |
| $y$ | 7 | 14 | 21 | 28 |

**b**

|     |   |    |    |    |
|-----|---|----|----|----|
| $x$ | 1 | 2  | 3  | 4  |
| $y$ | 9 | 10 | 11 | 12 |

**c**

|     |    |   |   |   |
|-----|----|---|---|---|
| $x$ | 1  | 2 | 3 | 4 |
| $y$ | -4 | 2 | 0 | 2 |

**d**

|     |               |               |               |   |
|-----|---------------|---------------|---------------|---|
| $x$ | 1             | 2             | 3             | 4 |
| $y$ | $\frac{1}{4}$ | $\frac{1}{2}$ | $\frac{3}{4}$ | 1 |

**e**

|     |   |    |    |    |
|-----|---|----|----|----|
| $x$ | 1 | 2  | 3  | 4  |
| $y$ | 6 | 11 | 16 | 21 |

**f**

|     |    |    |    |     |
|-----|----|----|----|-----|
| $x$ | 1  | 2  | 3  | 4   |
| $y$ | -3 | -6 | -9 | -12 |

**g**

|     |               |               |   |               |
|-----|---------------|---------------|---|---------------|
| $x$ | 1             | 2             | 3 | 4             |
| $y$ | $\frac{2}{3}$ | $\frac{4}{3}$ | 2 | $\frac{8}{3}$ |

**h**

|     |   |   |    |    |
|-----|---|---|----|----|
| $x$ | 1 | 2 | 3  | 4  |
| $y$ | 6 | 2 | -2 | -6 |



## Equations for relationships

9

a

|              |   |   |    |    |
|--------------|---|---|----|----|
| Bunches, $x$ | 1 | 2 | 3  | 4  |
| Bananas, $y$ | 4 | 8 | 12 | 16 |

b The number of bananas is equal to four times the number of bunches.

c  $y = 4x$

10

a

|            |    |     |     |     |
|------------|----|-----|-----|-----|
| Decks, $d$ | 1  | 2   | 3   | 4   |
| Cards, $c$ | 52 | 104 | 156 | 208 |

b The number of cards is equal to fifty-two times the number of decks.

c  $c = 52d$

11

a

|                       |       |       |       |       |       |
|-----------------------|-------|-------|-------|-------|-------|
| Week ( $W$ )          | 0     | 1     | 2     | 3     | 4     |
| Account total ( $A$ ) | \$300 | \$310 | \$320 | \$330 | \$340 |

b  $A = 300 + 10W$

12

a The number of marbles will be equal to four times the number of bags plus his original five.

b  $y = 4x + 5$